Module 9: Outlier Detection

- Basic concepts
- Statistical approaches
  - Parametric methods
  - Nonparametric methods
- Proximity-based approaches
- Reconstruction-based approaches
- Clustering and classification based approaches
- Mining contextual and collective outliers
- Outlier detection in high-dimensional data

General Idea

- The general idea behind statistical methods for outlier detection is to learn a generative model fitting the given data set, and then identify those objects in low-probability regions of the model as outliers
- A parametric method assumes that the normal data objects are generated by a parametric distribution with a finite number of parameters Θ
  - The probability density function of the parametric distribution $f(x, \Theta)$ gives the probability that object x is generated by the distribution
  - The smaller this value, the more likely x is an outlier
- A nonparametric method tries to determine the model from the input data

Detection of Univariate Outliers Based on Normal Distribution

- Assumption: Data are generated from a normal distribution
- Learn the parameters of the normal (i.e., Gaussian) distribution from the input data, and identify the points with low probability as outliers
- Example: suppose a city's average temperature values in July in the last 10 years are, in value-ascending order, 24.0°C, 28.9°C, 28.9°C, 29.0°C, 29.1°C, 29.1°C, 29.2°C, 29.2°C, 29.3°C, and 29.4°C
- A normal distribution is determined by two parameters: the mean, μ, and the standard deviation, σ
- Use the maximum likelihood method to estimate the parameters μ and σ

$$\ln L(\mu, \sigma^2) = \sum_{i=1}^{n} \ln f(x_i | (\mu, \sigma^2)) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln \sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

$$\hat{\mu} = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = 28.61, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 = 2.29$$

- 24.0°C is an outlier, since $L(24|(28.61, 2.29)) < 0.15\%$